

ARTESIA, NM, USA

WMO: 722676

Lat: 32.853N

Lon: 104.468W

Elev: 3541

StdP: 12.91

Time Zone: -7.00 (NAM)

Period: 99-19

WBAN: 03035

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	18.8	23.3	-1.7	5.7	40.7	1.1	6.6	45.6	33.6	57.7	29.0	55.9	6.1	160	0.537

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.3	101.7	65.2	99.5	65.8	97.3	66.0	73.1	86.1	72.0	85.6	71.0	84.9	11.8	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.4	127.9	75.3	69.6	124.3	74.9	67.7	116.2	74.5	39.2	86.1	38.1	84.9	37.1	84.2	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
28.9	25.2	21.0	DB	12.3	106.3	6.7	1.7	7.5	107.5	3.5	108.5	-0.2	109.4	-5.1	110.6
			WB	10.5	74.3	6.1	1.9	6.1	75.7	2.5	76.8	-0.9	77.8	-5.3	79.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.3	43.5	47.9	55.6	63.8	71.8	80.7	82.1	81.3	74.4	63.3	51.9	43.0	(6)
	DBStd	15.92	8.78	9.15	8.23	7.66	7.21	4.49	4.21	4.17	5.56	7.86	8.43	8.53	(7)
	HDD50	749	237	131	42	7	0	0	0	0	0	9	76	247	(8)
	HDD65	2792	666	480	303	110	26	0	0	0	5	122	397	683	(9)
	CDD50	5621	36	72	214	420	677	921	995	970	731	422	134	29	(10)
	CDD65	2189	0	1	10	73	238	471	530	505	286	70	5	0	(11)
	CDH74	28616	15	83	430	1405	3418	6338	6700	6058	3074	950	139	6	(12)
	CDH80	15204	1	11	104	537	1772	3761	3819	3346	1484	346	23	0	(13)
(14) Wind	WSAvg	9.3	8.5	9.8	10.5	11.7	10.9	10.4	8.5	7.6	8.2	8.5	8.4	8.4	(14)
(15) Precipitation	PrecAvg	12.5	0.4	0.4	0.6	0.6	1.3	1.4	1.9	1.5	1.7	1.1	0.6	0.6	(15)
	PrecMax	21.7	1.5	1.8	1.5	4.8	4.2	3.2	4.4	4.3	4.7	3.0	3.7	3.1	(16)
	PrecMin	7.9	0.0	0.0	0.1	0.0	0.1	0.0	0.3	0.0	0.2	0.0	0.0	0.0	(17)
	PrecStd	4.0	0.4	0.5	0.4	1.0	1.3	0.9	1.3	1.1	1.2	0.9	1.0	0.8	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.5	81.5	86.4	92.8	100.3	105.5	103.9	102.3	99.3	91.5	82.4	73.4	(19)
		MCWB	50.6	53.4	55.3	57.8	62.7	64.0	67.5	66.9	67.4	60.5	55.0	49.8	(20)
	2%	DB	71.7	76.5	82.4	89.9	97.2	102.0	100.2	99.4	95.0	88.2	78.7	69.8	(21)
		MCWB	49.4	51.1	53.5	56.5	61.0	63.4	67.5	67.6	65.6	61.5	54.5	49.0	(22)
	5%	DB	65.9	72.1	79.2	86.0	93.2	99.6	99.0	97.3	91.4	83.8	73.0	63.9	(23)
		MCWB	47.4	49.5	52.0	55.5	59.0	63.9	67.5	67.8	64.5	59.0	52.5	46.4	(24)
	10%	DB	61.0	66.2	74.7	82.3	90.4	97.0	95.3	94.7	89.7	80.6	68.2	59.0	(25)
		MCWB	44.2	47.2	50.6	54.5	59.0	64.2	67.8	67.8	64.9	59.0	50.9	44.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.2	56.0	61.7	65.0	70.0	74.3	74.4	74.1	72.8	70.4	62.7	54.4	(27)
		MCDB	69.4	73.1	71.3	75.9	85.1	87.4	89.2	85.5	82.4	76.8	69.0	64.4	(28)
	2%	WB	50.9	53.4	57.5	61.9	67.6	72.1	73.0	72.8	71.1	67.8	58.6	50.7	(29)
		MCDB	65.9	71.2	72.4	73.2	82.7	85.8	87.5	85.2	82.4	74.7	69.4	65.2	(30)
	5%	WB	48.3	51.0	55.0	59.4	65.7	70.4	71.8	71.6	69.6	65.2	55.1	48.0	(31)
		MCDB	63.2	68.7	71.4	75.7	80.6	85.2	86.7	85.0	81.1	74.6	67.6	62.3	(32)
	10%	WB	45.6	48.3	52.5	57.2	63.7	69.0	70.8	70.7	68.4	62.7	52.5	45.3	(33)
		MCDB	58.7	64.6	70.0	76.9	77.9	84.6	85.8	84.5	80.2	72.1	66.5	56.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	27.8	29.3	30.2	30.8	29.4	28.6	24.3	24.4	24.7	26.7	27.6	25.9	(35)
		MCDBR	35.7	37.6	37.7	35.9	34.6	33.5	28.9	28.4	29.8	33.6	34.3	34.0	(36)
	5% WB	MCWBR	19.1	19.0	17.7	16.1	14.6	11.7	8.6	8.4	10.3	14.0	17.0	18.1	(37)
		MCWBR	32.0	34.5	31.8	29.9	28.9	26.5	23.9	23.2	23.7	23.7	28.5	30.9	(38)
(40) Clear-Sky Solar Irradiance	taub		0.251	0.260	0.278	0.308	0.327	0.353	0.383	0.363	0.337	0.291	0.269	0.257	(40)
		taud	2.614	2.592	2.536	2.429	2.434	2.404	2.368	2.416	2.445	2.581	2.572	2.590	(41)
	Ebn at Noon		315	321	320	311	304	294	285	290	294	304	304	306	(42)
		Edh at Noon	24	27	31	36	36	38	39	36	34	27	25	23	(43)
(44) All-Sky Solar Radiation	RadAvg		1113	1413	1815	2228	2407	2458	2302	2103	1820	1482	1167	980	(44)
	RadStd		95	119	113	133	133	109	108	87	116	135	100	76	(45)

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only		N/A	N/A	N/A	N/A	-1.30	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page